Comprehensive List of Java Enterprise Edition (JEE) Interview Questions

1. Servlets

1.1 Q: What is a Servlet in Java?

A: A Servlet is a Java class used to handle requests and generate responses on a web server, providing dynamic content.

1.2 Q: How does the Servlet lifecycle work?

A: The Servlet lifecycle consists of initialization (init()), request handling (service()), and destruction (destroy()).

1.3 Q: What is the role of the service() method in a Servlet?

A: The service() method processes client requests. It is called every time a request is made to the servlet.

1.4 Q: Explain the init() method in a Servlet.

A: The init() method is called once when the servlet is loaded into memory, and it is used for one-time initialization tasks.

1.5 Q: What is the difference between doGet() and doPost()?

A: doGet() is used to handle HTTP GET requests (data retrieval), while doPost() is used for HTTP POST requests (data submission).

1.6 Q: What are Servlet containers?

A: Servlet containers are web servers or application servers that handle the lifecycle and execution of Servlets, e.g., Apache Tomcat.

1.7 Q: How can we share data between two Servlets?

A: Data can be shared using ServletContext, HttpSession, or request parameters.

1.8 Q: What is a filter in Servlet?

A: A filter is a Java class used to intercept and process requests and responses, often for tasks like authentication, logging, etc.

1.9 Q: What is the ServletConfig interface?

A: ServletConfig provides configuration parameters to a servlet, such as initialization parameters set in the web.xml.

1.10 Q: What is a session in Servlet?

A: A session in Servlet is used to store user-specific data across multiple requests, commonly implemented using cookies or URL rewriting.

2. JSP (Java Server Pages)

2.1 Q: What is JSP and how is it different from Servlets?

A: JSP is a view technology used for rendering dynamic content, while Servlets are used for handling requests and controlling the logic.

2.2 Q: What are the types of JSP tags?

A: The types of JSP tags include directives (<%@ %>), scriptlets (<% %>), expressions (<%= %>), and action tags (<jsp: %>).

2.3 Q: What are implicit objects in JSP?

A: Implicit objects are automatically available in JSP pages, including request, response, session, application, etc.

2.4 Q: How does JSP handle session management?

A: JSP handles sessions through the HttpSession object, which can store data across multiple requests from the same user.

2.5 Q: Explain the lifecycle of a JSP page.

A: The JSP lifecycle includes translation (converting to a Servlet), initialization (init()), request handling (service()), and destruction (destroy()).

2.6 Q: What are the advantages of JSP over Servlets?

A: JSP allows for separation of concerns (presentation and logic), simpler syntax, and reduces the amount of Java code in the page.

2.7 Q: How do you include one JSP page into another?

A: Use the include directive (<%@ include file="file.jsp" %>) or the jsp:include action tag (<jsp:include page="file.jsp" />).

2.8 Q: What is the forward() method in JSP?

A: The forward() method in JSP is used to forward a request from one servlet or JSP to another resource on the server.

2.9 Q: How does the session implicit object work in JSP?

A: The session object allows access to the HttpSession associated with the current user, enabling session management.

2.10 Q: What is the difference between include and forward in JSP?

A: include adds content from another resource into the current page, while forward redirects the request to another resource, terminating the current page's execution.

3. JDBC (Java Database Connectivity)

3.1 Q: What is JDBC and how does it work?

A: JDBC is an API that enables Java applications to interact with relational databases using SQL queries to retrieve and manipulate data.

3.2 Q: How do you establish a database connection using JDBC?

A: You establish a connection using DriverManager.getConnection() with the database URL, username, and password.

3.3 Q: What is the difference between Statement, PreparedStatement, and CallableStatement in JDBC?

A:

- Statement executes simple SQL queries.
- PreparedStatement is precompiled, preventing SQL injection and improving performance.
- CallableStatement is used to execute stored procedures.

3.4 Q: What is ResultSet in JDBC?

A: ResultSet represents the result of a query, which can be used to retrieve data from the database row by row.

3.5 Q: What is a Connection Pool?

A: A connection pool is a collection of database connections that are reused to optimize database connectivity performance.

3.6 Q: What are the different types of JDBC drivers?

A: The four types of JDBC drivers are:

1. Type-1 (JDBC-ODBC Bridge)
2. Type-2 (Native-API)
3. Type-3 (Network Protocol)
4. Type-4 (Thin Driver)

3.7 Q: What are the commit() and rollback() methods in JDBC?

A: commit() commits the transaction, making changes permanent. rollback() undoes the changes made during the current transaction.

3.8 Q: What is the use of setAutoCommit() in JDBC?

A: setAutoCommit() controls whether each individual statement is committed automatically or not. By default, it is true.

3.9 Q: How do you handle exceptions in JDBC?

A: JDBC exceptions are handled using try-catch blocks, and specific SQLException objects are used to capture detailed error information.

3.10 Q: What is batch processing in JDBC?

A: Batch processing allows multiple SQL statements to be executed as a single batch, improving performance by reducing the number of database round trips.